A Euclidean Affinity-Augmented Hyperbolic Neural Network for Semantic Segmentation of Remote Sensing Images

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***Abstract*—Semantic segmentation of remote sensing images (RSIs) plays a pivotal role in advancing geospatial analyses and applications across diverse fields, such as urban planning and environmental monitoring. Traditional learning paradigms predominantly utilize Euclidean spaces for feature extraction. This approach can introduce spatial distortions when representing objects, as Euclidean architectures typically focus on locality and are optimized for grid data, not always yielding optimal geometrical representations for data structured in non-Euclidean spaces. To address these problems, we propose EAAHNet, the first fully hyperbolic neural network designed for the semantic segmentation of RSIs. EAAHNet employs the Lorentz model to reformalize conventional Euclidean-based neural network operations, ensuring the preservation of hyperbolic properties. Furthermore, to account for the inherently Euclidean nature of ground objects, we propose a Euclidean affinity-augmented hyperbolic attention module (EAAHAM) that enriches contextual dependencies through an attention fusion manner. This enhancement significantly improves the network's capacity to discern pixel-wise semantics. Extensive experiments conducted on the ISPRS Vaihingen, ISPRS Potsdam, and LoveDA datasets demonstrate EAAHNet's superior performance over several state-of-the-art methods. Additionally, the ablation study verifies the impacts of EAAHAM.**

***Index Terms*—Attention bias, contextual affinity, remote sensing images (RSIs), semantic segmentation, synergistic attention.**

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I. Introduction

SEMANTIC segmentation, a key task in computer vision, involves labeling each pixel in an image, thereby performing classification at the pixel level [1], [2], [3]. This process is essential for various applications, such as lane analysis in self-driving cars [4]. Unlike traditional image predictions, semantic segmentation provides detailed maps of objects, capturing their precise locations and shapes [5], [6]. As segmentation technologies evolve, they have been increasingly applied to the complex, data-intensive challenges of remote sensing [7], [8], [9]. Applications in this field range from environmental monitoring and agricultural land cover classification to identifying tree species and analyzing urban buildings and land use [10], [11], [12], [13], [14].

Remote sensing images (RSIs), which contain detailed information about ground objects, often face issues like imbalanced category distribution [15], [16], [17], [18]. Additionally, variations in shapes, colors, and textures of ground objects create significant intra-class differences and minimal inter-class differences [19]. These factors pose major challenges for semantic segmentation of RSIs [20], [21]. Initially, RSI semantic segmentation relied on methods using hand-crafted features [22], [23]. However, as RSI resolution improves, traditional methods struggle to accurately extract features and achieve precise segmentation. These methods often require extensive domain-specific knowledge and manual feature design, making them less adaptable and increasing development costs [24]. Recently, deep learning methods, particularly convolutional neural networks (CNNs) [25], have inspired numerous advanced segmentation networks, demonstrating the flexibility and effectiveness of CNN-based models [26], [27], [28]. However, the fixed geometric structures and local receptive fields of CNNs limit their ability to capture long-range context [29], [30], [31].

Leveraging contextual information has been proven to be useful in boosting the discriminability of learned representations [3], [32]. Two primary strategies have been used to improve contextual integration. The first involves multiscale dilated convolutions or pooling functions to capture features at different scales. For example, MLCRNet [33] demonstrated excellent performance on the ISPRS Potsdam [34] and Vaihingen [35] benchmarks by aggregating multilevel context. Similarly, Du et al. [36] created a network aimed at mapping urban functional zones.

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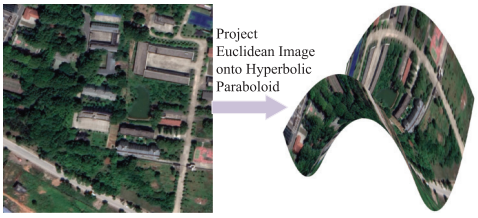


Fig. 1. Illustration of projecting image onto hyperbolic paraboloid.

The second strategy involves integrating attention modules to capture long-range dependencies [37], [38]. For instance, Li et al. [39] introduced dual attention and deep fusion strategies to tackle the challenges of segmenting large-scale satellite imagery. Li et al. [40] designed a multiattention network that effectively balances contextual dependency extraction and computational efficiency. HCANet [41] combines cross-level contextual and attentive representations, and EDENet [41] uses an attention module to adaptively incorporate edge information. HMANet [42] captures spatial, channel, and category correlations, and Ding et al. [43] introduced LANet [43]. Overall, the attention mechanism has proven to be exceptionally beneficial in RSI applications, aiding models in managing the subtle nuances within and between object classes [21], [44], [45], [46].

Contemporary models predominantly learn from data in Euclidean space, characterized by its flat structure and straightforward operations like vector addition, inner product, and averaging. This approach simplifies the representation of raw data into low-dimensional latent embeddings. However, neural architectures designed for Euclidean space often focus on local features and are initially crafted for grid data, which may not offer the most effective geometric representations for structured data in non-Euclidean spaces [47], [48], [49], [50]. Heimann et al. [51] explored the surface's geometry property and incorporated an orthogonal discrete cosine transform basis function to strategically insert additional points. Specifically, consideration should be given to non-Euclidean spaces such as elliptic space, characterized by constant positive curvature, and hyperbolic space (see example in Fig. 1), known for its constant negative curvature [52].

As illustrated in Fig. 1, projecting Euclidean image features onto a hyperbolic paraboloid enables an exponential expansion of space, effectively capturing both fine-grained and coarse-grained structural dependencies. This is crucial in RSIs, where features like roads, rivers, and buildings form nested and interconnected spatial systems. Unlike elliptic spaces, which are positively curved and suitable for compact, closed representations (e.g., spherical signals), hyperbolic space, with its constant negative curvature, better models the tree-like, branching nature of spatial hierarchies. This aligns with findings from prior work [52], [53], [54], which highlight the advantages of non-Euclidean embeddings in tasks requiring global awareness.

By embedding feature vectors onto a Lorentzian hyperboloid, our network benefits from preserved geodesic distances, meaning relationships between distant but semantically related regions (e.g., buildings across a city block) are maintained without contraction or distortion. Moreover, hyperbolic convolution and attention operations further ensure that contextual information propagates efficiently across layers, supporting deeper semantic reasoning. In this sense, the hyperbolic manifold serves not only as a computational substrate but also as a principled inductive bias for learning representations that mirror the geometric complexity of the real world as captured in RSIs.

Beyond structured data, hyperbolic space can also articulate intrinsic relationships and developmental progressions in regular samples. From a cognitive science standpoint, it is known that humans use hierarchical structures to organize actions and categorize objects [55]. Notably, Zhou et al. [56] revealed that both natural odors and human sensory descriptions of scents are best described in a 3-D hyperbolic space. Interestingly, recent findings [57] indicate that features of datasets like miniImageNet and CIFAR, learned in Euclidean space using architectures such as VGG, Inception, and ResNet, exhibit hyperbolic traits. Thus, there is a compelling argument for the development of hyperbolic deep neural networks to handle the complexities of irregular data more effectively, enhance the understanding and reasoning of complex relationships beyond Euclidean geometry, and achieve breakthroughs in hyperbolic space applications [58].

Therefore, there are two main issues to be addressed while segmenting RSIs in hyperbolic space.

1) Extracting visual features from images and then transforming into a vectorized representation in Euclidean space has been the de facto dominating solution of vision tasks. However, Euclidean is a flat embedding space, leading to spatial distortions when representing real-world stuff. Inherently, the neural architectures operating in the Euclidean space rely heavily on locality and are originally designed for grid data, thus not necessarily providing the most powerful or meaningful geometrical representations for structured data in a non-Euclidean space.
2) Moreover, a fully hyperbolic network lacks a complete understanding of hierarchy-agnostic features of RSIs, like texture and color. By combining hyperbolic and Euclidean representations, massive projections of logarithmic and exponential maps are required. Therefore, it is necessary to complement the hyperbolic context with Euclidean counterpart in a simple yet effective manner.

In this article, we propose a Euclidean affinity-augmented hyperbolic neural network for semantic segmentation of RSIs (EAAHNet), which learns and refines representations in hyperbolic space. Specifically, we generalize the CNN operators in hyperbolic space by adopting the Lorentz model, which is an isometric model insusceptible to numerical instability. Moreover, regarding the non-hierarchical properties of ground objects in RSIs, we propose to embed a Euclidean affinity-augmented hyperbolic attention module (EAAHAM) to complete contextual dependencies with an attention fusion design. The main contributions are as follows.

1) We propose the first fully hyperbolic neural network for semantic segmentation of RSIs, termed as EAAHNet,

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inheriting the encoder–decoder architecture. Specifically, we generalize the convolution layer, batch normalization, nonlinear activation, residual connection, Softmax function, and deconvolution layer in hyperbolic space for learning low-distortion representations of RSIs. Regarding that Poincaré ball is more susceptible to numerical instability, we adopt the Lorentz model as the isometric model of our hyperbolic network.

2) Moreover, we propose EAAHAM that models pairwise correlations in both hyperbolic and Euclidean spaces simultaneously in a parallel manner. Followed by a weighted summation of hyperbolic attention map and Euclidean attention map, we have a well-rounded context-included attention weights within one matrix. Aggregated with input hyperbolic feature, the EAAHAM-refined feature is produced with a comprehensive context from two space domains.
3) Extensive experiments conducted on three representative benchmarks, ISPRS Vaihingen, ISPRS Potsdam, and LoveDA, have validated the efficacy and superiority of the proposed network. Moreover, an ablation study examines the effects of embedding EAAHAM.

The structure of this article is outlined as follows. Section II reviews the literature on semantic segmentation of RSIs and learning in hyperbolic spaces. Section III details our proposed EAAHNet. Section IV discusses the experimental setup, results, and a critical analysis of model performances. Section V concludes this article with a summary of the research findings.

II. Related Works

A. Semantic Segmentation of RSIs

CNNs remain a cornerstone for RSI segmentation due to their strong capabilities in hierarchical spatial feature extraction [2]. Early frameworks such as FCN [26], U-Net [28], and SegNet [27] established foundational architectures. These were subsequently enhanced by methods like ResUNet-a [59], which incorporates residual connections and attention mechanisms to improve multiscale representation, and LANet [43], which introduces patch-wise attention to better capture fine-grained details. SDFCNv2 [60] and MASA-SegNet [61] further leverage dual-branch and frequency-aware modules to handle complex spatial variations. Hybrid attention models like HMANet [42], Li et al. [39], and dual-structure networks such as WiCoNet [62] emphasize both spatial and channel feature enhancement to deal with occlusion and category ambiguity in cluttered urban environments.

Advances in deep learning have significantly enhanced the semantic segmentation of RSIs. Various works, including [63], [64], have successfully adapted general semantic segmentation techniques to these images, yielding satisfactory outcomes. Given the rich spatial details present in RSIs compared to natural scene imagery, numerous studies have focused on developing specialized deep learning architectures [65], [66], [67]. For instance, Liu et al. [68] crafted an adaptive fusion network that selectively concentrates on differently sized objects across multiple feature levels. Yang et al. [37] implemented a strategy involving a multipath encoder and an attention-fused framework to facilitate concurrent fusion of multipath and multilevel features. Li et al. [40] devised a multiattention scheme that hierarchically aggregates contextually relevant features with linear complexity. With the introduction of frequential features, SSCBNet integrates synergetic attention layers and cross-fusion modules to harness both spectral and spatial information, addressing the intrinsic complexities of urban and natural landscapes within RSIs [69]. More recently, BEDSN [70] introduces a boundary-enhanced dual-stream network that explicitly models boundary cues to improve contour precision and intra-class consistency, leveraging edge-aware modules and cross-stream interactions. UNetMamba [71] proposes a Mamba-based segmentation decoder and a local supervision module in a lightweight U-Net-like framework, achieving a strong balance between accuracy and efficiency for high-resolution images. Likewise, RS3Mamba [72] further enhances the modeling of global semantic dependencies by introducing a dual-branch architecture built upon visual state space blocks and a novel cross-branch fusion strategy.

Despite these advancements, many existing methods are fundamentally grounded in Euclidean geometry, which limits their capacity to model the hierarchical and nonlinear spatial relationships prevalent in RSIs. Objects in remote sensing imagery—such as buildings, roads, and vegetation—are often embedded within nested spatial structures that are better represented in non-Euclidean geometries. Hyperbolic space offers a mathematically natural framework for embedding such hierarchies, enabling richer relational modeling with lower distortion. While prior works have primarily focused on spatial, frequency, or attention mechanisms, fewer efforts have explored leveraging hyperbolic geometry to encode multiscale and nested semantic structures. Incorporating hyperbolic representation allows models to preserve both local details and global hierarchies with geometric fidelity. This opens new avenues for designing neural architectures that can more effectively handle the topological complexity inherent in RSIs.

B. Learning in Hyperbolic Space

Learning representations in hyperbolic space has gained significant attention for its ability to capture hierarchical and tree-like structures in data [73], [74]. Nickel and Kiela [75] pioneered the use of hyperbolic space for learning hierarchical representations, especially through the Poincaré ball model. Their research shows that hyperbolic embeddings can greatly outperform Euclidean embeddings in tasks like link prediction in hierarchical datasets. Hyperbolic spaces can be visualized as continuous versions of trees, making them well-suited for representing hierarchical structures. As illustrated in Fig. 2, embedding hierarchical structures in Euclidean space leads to distortions, especially as we move further from the root of the tree. Euclidean space expands linearly, making it difficult to accurately capture the increasing number of relationships and branches deeper in the hierarchy. Conversely, hyperbolic space expands exponentially, allowing it to naturally represent hierarchical data with minimal distortion. This exponential expansion ensures that distances between nodes reflect their

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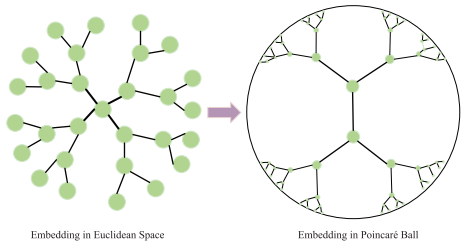


Fig. 2. Brief comparison between embedding of trees in Euclidean space (left) and hyperbolic space (right). Notably, we visualize the distance with Poincaré ball model for an intuitive display.

hierarchical relationships more accurately. For example, a hyperbolic embedding would place a central node (e.g., urban areas) at the origin and spread out other categories (e.g., buildings, parks, roads) in a way that preserves their hierarchical and spatial relationships more effectively.

Following the foundational models, several studies explored the use of hyperbolic spaces for graph embeddings. Chami et al. [76] extended hyperbolic embeddings to more complex graph structures, introducing the hyperbolic graph neural network, which adapts the graph neural network framework to operate in hyperbolic space. Moreover, hierarchical structures, similar to those observed in natural language processing and graph data, are also present in images. Khrulkov et al. [77] detected significant hyperbolicity in image dataset embeddings, indicating a strong inherent tree structure. This hyperbolic nature extends beyond final embeddings, affecting intermediate layers as well. Recent studies have started to incorporate hyperbolic geometry into vision tasks, focusing on the Poincaré ball and the Lorentz model to describe hyperbolic spaces and adapt neural network layers to these geometries. These adaptations pose significant challenges due to the complex nature of defining hyperbolic equivalents for operations such as addition, multiplication, and statistical analysis [49], [78].

The combination of Euclidean and hyperbolic spaces has been explored in previous works, such as [21], where the focus was primarily on embedding Euclidean features into hyperbolic space to leverage hierarchical data structures. Our approach differs significantly in two key aspects. First, we use a parallel architecture where features are learned independently in both Euclidean and hyperbolic spaces, allowing for a richer and more flexible representation of both flat and hierarchical structures inherent in RSIs. Second, our EAAHAM introduces a novel fusion mechanism that dynamically adjusts the contribution from each geometric space based on the learned context, which ensures optimal feature fusion. This stands in contrast to existing methods that rely on fixed embeddings and static fusion strategies, limiting their ability to adapt to varying data complexities. By doing so, we not only address the heterogeneity between the two spaces but also enhance the overall robustness and accuracy of the model in complex segmentation tasks.

III. Method

A. Overview

As illustrated in Fig. 3, our proposed EAAHNet is a fully hyperbolic neural network tailored for the semantic segmentation of RSIs. The model adopts a symmetric encoder–decoder structure, where all operations are conducted in the Lorentz model of hyperbolic space.

The pipeline begins by embedding the input image—originally defined in Euclidean space—into a Lorentzian hyperbolic space L_K^n. This embedding is performed via a logarithmic map, which projects each pixel feature vector onto the hyperboloid manifold while preserving angular and structural relationships. We denote the hyperbolic feature at spatial position (h, w) as $x_{h,w} \in L_K^n$.

Once initialized in hyperbolic space, the data is processed through a sequence of Lorentz convolution layers (LConv), hyperbolic encoder blocks (HEBs), and decoder blocks (HDBs), as shown in Fig. 3(a). Each HEB and HDB is composed of Lorentz-compatible operations, including Lorentz batch normalization (LBN), Lorentz ReLU (LNA), Lorentz deconvolution (LDeConv), and our proposed EAAHAM. The detailed operation flow of HEBs and HDBs is visualized in Fig. 3(b) and (c), respectively.

Notably, EAAHNet performs all attention, normalization, and activation directly within the hyperbolic manifold, with special care taken to preserve curvature constraints and maintain geometric validity. At the final decoder stage, a Lorentz convolution followed by Lorentz Softmax (L-Softmax) is applied to assign pixel-wise semantic labels by comparing features to class prototypes using Lorentzian distances.

EAAHNet is designed to exploit the representational advantages of hyperbolic space—particularly its capacity to model hierarchical and fine-grained structures—while retaining compatibility with the global layout of remote sensing imagery. To further enhance geometric expressiveness, EAAHAM fuses contextual dependencies from both hyperbolic and projected Euclidean affinity spaces.

B. Lorentz Model

The Lorentz model, also known as the hyperboloid model, is a pseudo-Riemannian representation of hyperbolic space embedded in Minkowski spacetime. It has been widely used in hyperbolic deep learning due to its superior numerical stability, efficient distance computations, and closed-form geodesics, making it especially advantageous for modeling the complex hierarchical structures often found in RSIs.

The n-dimensional hyperbolic space $\mathbb{H}_K^n$ is defined as a Riemannian manifold (M^n, g_x^K) with constant negative curvature $K < 0$, where g_x^K is the Lorentzian metric at point x. Specifically, the Lorentz model describes this space as the upper sheet of a two-sheeted hyperboloid embedded in an $(n+1)$-dimensional Minkowski space

$$L_K^n = \left\{ x \in \mathbb{R}^{n+1} \mid \langle x, x \rangle_L = \frac{1}{K},\ x_t > 0 \right\} \tag{1}$$

where $x = [x_t, x_s]^T$, with x_t denoting the time-like component and $x_s \in \mathbb{R}^n$ the spatial components. The Lorentzian inner product $\langle \cdot, \cdot \rangle_L$ is given by

$$\langle x, y \rangle_L := -x_t y_t + x_s^T y_s = x^T \cdot \mathrm{diag}\,(-1, 1, \ldots, 1) \cdot y. \tag{2}$$

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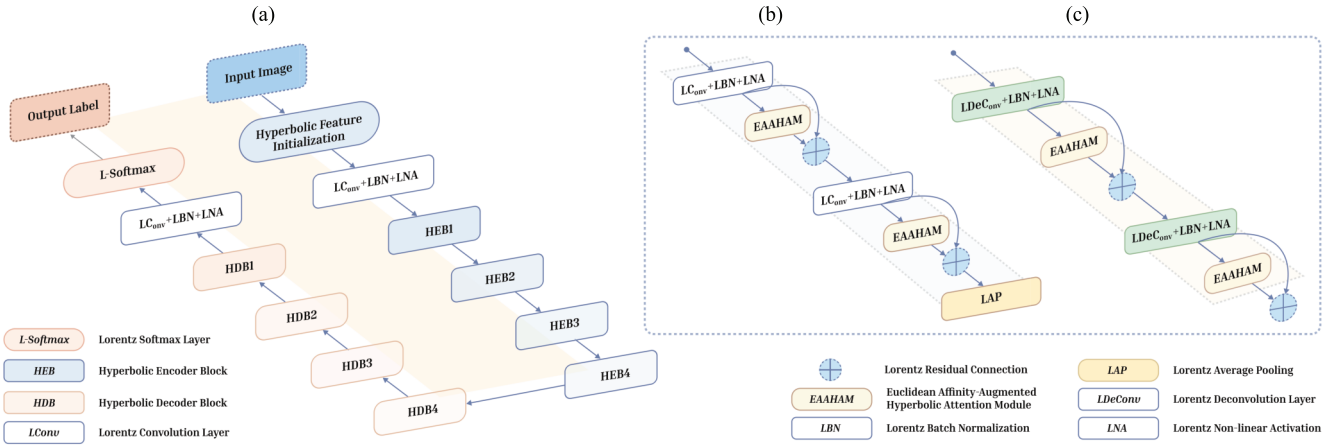


Fig. 3. Details of EAAHNet. (a) Framework of EAAHNet. (b) Pipeline of HEB. (c) Pipeline of HDB.

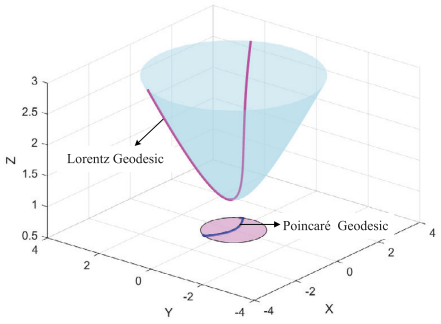


Fig. 4. Geodesics in Lorentz and Poincaré models: the Lorentz model projects hyperbolic space onto the upper hyperboloid in Minkowski space. Geodesics in Lorentz space correspond to hyperbolic straight lines defined by intersecting planes through the origin.

This model guarantees that each point $x \in L_K^n$ lies on the hyperboloid defined by the constraint $\langle x, x\rangle_L = (1/K)$. The natural origin of this space is defined as $\bar{0} = [((1/-K))^{1/2}, 0, \ldots, 0]^T$.

Fig. 4 visualizes how geodesics are constructed in the Lorentz model compared to the Poincaré model. Notably, geodesics in the Lorentz model correspond to intersections of the hyperboloid with planes through the origin, offering exact formulas for exponential and logarithmic maps. This makes the Lorentz model highly suitable for Riemannian optimization in deep learning tasks.

For RSIs, whose content often reflects hierarchical spatial organization (e.g., roads, blocks, buildings, and green spaces), the Lorentz model's ability to preserve both local and global geometrical relationships with minimal distortion makes it a powerful tool for feature learning in non-Euclidean space.

Compared with the Poincaré ball, the Lorentz model offers numerically stable computations and more consistent distance measurements. It allows for the direct construction of deep networks through hyperbolic layers and attention mechanisms, which we extend in our architecture to address the geometric complexity of RS imagery.

C. Lorentz-Based Hyperbolic Components

Extending from Euclidean spaces, we have reformulated Lorentz-based hyperbolic components for integration into any neural network architecture.

1) Hyperbolic Feature Initialization: To enable neural processing within the Lorentz model of hyperbolic space, the raw Euclidean image features must first be embedded onto the hyperboloid manifold. This transformation is achieved through the logarithmic map, a standard tool for projecting Euclidean vectors into hyperbolic space in a geometry-consistent manner.

Given an input pixel-level feature $x \in \mathbb{R}^n$ extracted from the image, we lift it to the Lorentzian manifold $L_K^n \subset \mathbb{R}^{n+1}$ using the logarithmic map defined by

$$\phi(x) = \left[\sqrt{\|x\|^2 + \frac{1}{K}},\, x^T\right]^T \tag{3}$$

where $K > 0$ is the absolute value of the constant negative curvature of the hyperbolic space. This ensures that $\phi(x)$ lies on the upper sheet of the two-sheeted hyperboloid

$$L_K^n = \left\{y \in \mathbb{R}^{n+1} \mid \langle y, y\rangle_L = -\frac{1}{K},\ y_0 > 0\right\} \tag{4}$$

with the Lorentz inner product defined as

$$\langle u, v\rangle_L = -u_0 v_0 + \sum_{i=1}^{n} u_i v_i. \tag{5}$$

This initialization ensures that all subsequent operations—convolution, normalization, and attention—operate within a valid hyperbolic geometry. Furthermore, it preserves the original spatial and semantic relationships of the input features by mapping their relative positions onto the hyperbolic manifold without distortion of angular hierarchy.

In our implementation, the logarithmic mapping is applied independently to each pixel feature vector at the input layer. The resulting tensor $\phi(X) \in \mathbb{R}^{H\times W\times(n+1)}$ then serves as the hyperbolic input to the first LConv. This step is crucial for ensuring geometric consistency throughout the EAAHNet architecture.

2) Lorentz Convolution Layer: Convolution in hyperbolic space requires rethinking the fundamental operations used in standard Euclidean convolution. In the Lorentz model, which represents hyperbolic geometry using a Minkowski inner product, the Euclidean convolution operation must be replaced with a geometry-aware transformation that respects curvature and manifold constraints. Specifically, all feature

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vectors must lie on the hyperboloid defined by $\langle x, x\rangle_L = -1/K$, where $\langle\cdot,\cdot\rangle_L$ denotes the Lorentzian inner product and K is the curvature.

Given an input feature map $x = \{x_{h,w} \in L_K^n\}_{h=1,w=1}^{H,W}$ in Lorentz space, we extract a local patch centered at position (h', w') using a window of size $[\tilde{H}, \tilde{W}]$. Each local patch is denoted as $x_{cf} = \{x_{h'+\delta\tilde{h},w'+\delta\tilde{w}} \in L_K^n\}$. Unlike Euclidean CNNs that apply linear operations directly on these vectors, we first use a Lorentzian concatenation operation (LCAT) to aggregate the patch into a single representation. The LCAT operator maintains the manifold constraint and outputs a new point in $L_K^{n\tilde{H}\tilde{W}}$

$$y = \mathrm{LFC}\left(\mathrm{LCAT}\left(\left\{x_{cf}\right\}\right)\right). \tag{6}$$

Here, LFC is the Lorentz fully connected layer, defined as

$$y = \mathrm{LFC}(x) = \begin{bmatrix} \sqrt{\|\psi(Mx+b)\|^2 - \frac{1}{K}} \\ \psi(Mx+b) \end{bmatrix} \tag{7}$$

where ψ is a nonlinearity (e.g., tanh or Lorentz ReLU), and M and b are learnable weights and biases, respectively. The spatial component is preserved while the time component is recalculated to ensure hyperbolicity. This operation projects the result back onto the hyperboloid.

The overall Lorentz convolution process can be interpreted as a kernel-weighted aggregation of manifold-aware embeddings that maintains consistency with the negative curvature of the space. This design allows EAAHNet to encode local spatial dependencies while preserving the underlying geometry.

Additionally, to stabilize learning, the convolutional result is often followed by LBN and Lorentz nonlinear activation (LReLU), maintaining the flow of hyperbolic-compatible signals throughout the network. The use of LCAT before convolution is crucial since concatenating vectors in hyperbolic space is non-trivial, and the transformation needs to retain the time-like structure required by Lorentzian manifolds.

3) Lorentz Batch Normalization: Batch normalization plays a crucial role in accelerating convergence and stabilizing training in deep neural networks. In the context of hyperbolic neural networks, however, conventional Euclidean batch normalization fails to preserve the intrinsic geometry. We thus extend batch normalization into Lorentz space by designing LBN, which operates natively in the hyperboloid model.

LBN begins by computing a batch-wise mean in Lorentz space. Unlike the Euclidean case, where the arithmetic mean suffices, we compute the Lorentzian centroid—a mean under the pseudo-Riemannian metric. Given a batch of hyperbolic vectors $\{x_i\}_{i=1}^m \subset L_K^n$, the centroid μ is computed as

$$\mu = \frac{\sum_{i=1}^m v_i x_i}{\sqrt{-K\left\|\sum_{i=1}^m v_i x_i\right\|^2}} \tag{8}$$

where v_i is a weighting factor (typically uniform), and K denotes the negative curvature. This operation ensures that $\mu \in L_K^n$.

To normalize, we shift features using Lorentz-compatible parallel transport. Let $\mathrm{PT}(x_i, \mu)$ denote this operation, defined simply as

$$\mathrm{PT}(x_i, \mu) = x_i - \mu \tag{9}$$

which recenters features around the hyperbolic origin without distorting their geometric properties.

Next, we compute the hyperbolic variance using the Fréchet formulation based on the Lorentzian distance $d_L(x_i, \mu)$

$$\sigma^2 = \frac{1}{m}\sum_{i=1}^m d_L^2(x_i, \mu). \tag{10}$$

Finally, normalized features are obtained as

$$y_i = \frac{\mathrm{PT}(x_i, \mu)}{\sigma} \tag{11}$$

ensuring that the scaled representations lie on the hyperboloid. This entire process guarantees that the normalization respects the pseudo-Riemannian structure of the Lorentz model, offering numerically stable and geometrically consistent training dynamics.

4) Lorentz Nonlinear Activation: Nonlinear activation functions are critical in deep neural networks as they enable the modeling of complex, nonlinear relationships in the data. However, standard activations such as ReLU are formulated under Euclidean assumptions, making them incompatible with the pseudo-Riemannian geometry of Lorentzian hyperbolic space.

To address this, we adopt Lorentz ReLU (LReLU), a geometry-preserving activation function specifically designed for the Lorentz model. It applies ReLU only to the spatial components of the hyperbolic vector while recalculating the time-like component to ensure the output vector remains on the hyperboloid defined by the Lorentzian constraint.

Let $x = [x_0, x_s]^T \in L_K^n$ be a hyperbolic feature vector, where x_0 is the time component and $x_s \in \mathbb{R}^n$ represents the spatial components. LReLU is defined as

$$y = \mathrm{LReLU}(x) = \begin{bmatrix} \sqrt{\|\mathrm{ReLU}(x_s)\|^2 - \frac{1}{K}} \\ \mathrm{ReLU}(x_s) \end{bmatrix} \tag{12}$$

where $\|\cdot\|$ denotes the Euclidean norm and K is the negative curvature. ReLU is applied componentwise on x_s, setting all negative values to zero. The new time component is rederived from the updated spatial norm to ensure that $y \in L_K^n$.

This formulation ensures that the nonlinearity does not project the feature outside the hyperbolic manifold, thus preserving the manifold's intrinsic geometry. By using LReLU, we enable deep architectures to benefit from expressive nonlinear transformations while maintaining Lorentzian consistency.

5) Lorentz Residual Connection: Residual connections play a pivotal role in stabilizing and accelerating the training of deep neural networks by facilitating gradient flow and preserving low-level information. In the context of hyperbolic geometry, however, traditional Euclidean addition violates manifold constraints. To resolve this, we employ Lorentz residual connections (LRCs), which leverage the Lorentz addition operation to ensure the result remains on the hyperbolic manifold.

Given an input feature vector $x \in L_K^n$ and its transformation $f(x)$ by a sequence of Lorentz-compatible operations such as LBN (11), Lorentz ReLU (12), and LFC layers (7), the output of LRC is computed as

$$y = x + \mathrm{LADD}(f(x), x) \tag{13}$$

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where LADD is a Lorentz addition operator that combines two hyperbolic vectors while preserving the Lorentz manifold structure. It is defined as

$$\mathrm{LADD}(a,b) = a + b - \frac{\langle a,b\rangle_L}{1+\langle a,\xi\rangle_L}\xi \tag{14}$$

where $\langle\cdot,\cdot\rangle_L$ denotes the Lorentz inner product, and ξ is a reference vector on the hyperboloid, often chosen as the origin or the base point for parallel transport.

The correction term involving $\langle a,b\rangle_L$ and ξ ensures that the composite output remains valid in the Lorentzian space and compensates for curvature-induced distortion. This design allows residual learning in hyperbolic geometry to retain the benefits of identity mapping while adhering to the non-Euclidean manifold structure.

In the proposed EAAHNet, LRCs facilitate deep signal propagation across hyperbolic layers, enhancing geometric consistency and stability during semantic segmentation of RS images, especially for complex urban layouts where feature reuse is critical.

6) Lorentz Softmax Function: The softmax function is fundamental in classification tasks as it converts raw prediction scores into a probability distribution over classes. However, its standard Euclidean formulation is not directly applicable in hyperbolic space. To address this, we introduce the Lorentz Softmax (L-Softmax), which extends the classical softmax into the Lorentz model.

In our formulation, the similarity between a hyperbolic feature vector x and the class prototype μ_c is computed based on the Lorentz distance $d_L(x,\mu_c)$. Then, the probability that x belongs to class c is defined as

$$S(x)_c = \frac{e^{-d_L(x,\mu_c)}}{\sum_{j=1}^{C} e^{-d_L(x,\mu_j)}}. \tag{15}$$

Here, μ_c denotes the class centroid in hyperbolic space, obtained by minimizing the Lorentzian variance across samples of class c, while C is the total number of classes. The use of the negative Lorentz distance in the exponent ensures that closer points (more similar) yield higher probability, just as in Euclidean softmax. The Lorentz distance metric d_L inherently encodes the curvature of the space, making the probability assignment geometrically consistent.

7) Lorentz Average Pooling: Lorentz average pooling (LAP) is specifically tailored to compute the average of features within hyperbolic space, respecting its geometric constraints. Unlike Euclidean average pooling, which simply calculates the arithmetic mean, LAP seeks the Karcher (or Fréchet) mean—the point on the manifold that minimizes the sum of squared Lorentz distances to all feature points in the set.

Given a set of hyperbolic feature vectors $\{x_i \in L_K^n\}_{i=1}^{m_p}$, the pooled feature μ_f is defined as

$$\mu_f = \arg\min_{y\in L_K^n} \sum_{i=1}^{m_p} d_L^2(y,x_i) \tag{16}$$

where $d_L(\cdot,\cdot)$ denotes the Lorentz distance, and μ_f lies on the Lorentz hyperboloid. This optimization is solved iteratively using gradient descent methods compatible with hyperbolic geometry. By maintaining the pooling result within the manifold, LAP ensures the consistency and validity of downstream hyperbolic operations, supporting stable and meaningful aggregation in feature downsampling stages.

8) LDeConv Layer: The LDeConv layer plays a crucial role in the decoder phase of hyperbolic neural networks, facilitating the upsampling process necessary for dense prediction tasks such as semantic segmentation.

In contrast to traditional Euclidean deconvolution that utilizes standard grid-based transposed convolution, LDeConv operates within the Lorentzian hyperbolic space, requiring the output features to remain on the hyperboloid. To achieve this, the deconvolution kernel is first applied in a local Euclidean tangent space, and then the result is projected back onto the Lorentz manifold. This is done using exponential and logarithmic maps that transition between hyperbolic and tangent space representations.

Let the input feature be $\mathbf{x} \in L_K^n$ and the deconvolution kernel be $K_{\mathrm{dec}} \in \mathbb{R}^{m\times n\times \tilde{H}\times \tilde{W}}$. The process begins by mapping $\mathbf{x}$ to the tangent space at the origin via the logarithmic map $\log_o(\mathbf{x})$. A standard Euclidean transposed convolution is applied in this space, followed by projection back to the hyperbolic space via the exponential map $\exp_o$. This ensures that the output remains consistent with hyperbolic geometry

$$\mathbf{y}_{h,w} = \exp_o\left(\mathrm{DeConv}\left(\log_o\left(\mathbf{x}_{cf}\right), K_{\mathrm{dec}}\right)\right) \tag{17}$$

where $\mathbf{x}_{cf}$ denotes the receptive field around position (h,w), and DeConv is the standard transposed convolution operation.

Furthermore, to preserve boundary semantics, hyperbolic-aware padding is introduced. It ensures that padding values do not break the manifold constraints and are drawn from points near the hyperbolic origin or according to the manifold curvature K. This consideration is critical for spatial consistency in feature reconstruction.

Overall, LDeConv enhances EAAHNet's ability to upscale feature maps without distorting their geometric structure, allowing richer spatial recovery in hyperbolic space.

9) Hyperbolic Softmax Cross-Entropy Loss: In our framework, classification is conducted in hyperbolic space using the Lorentz model, and we adopt the Hyperbolic Softmax Cross-Entropy as the loss function. This formulation ensures compatibility with the non-Euclidean geometry of hyperbolic space, allowing gradients to propagate meaningfully along the manifold.

Given a hyperbolic feature vector $x \in L_K^n$ and the centroid μ_c of class c, the predicted class probability is computed using the Lorentzian softmax (15). Then, the Hyperbolic Softmax Cross-Entropy Loss for a labeled sample (x,y) is defined as

$$\mathcal{L}_{\mathrm{hyp}} = -\log\left(\frac{e^{-d_L(x,\mu_y)}}{\sum_{j=1}^{C} e^{-d_L(x,\mu_j)}}\right) \tag{18}$$

where y denotes the ground-truth label. This formulation ensures that samples are pulled closer to their class centroids on the hyperboloid while pushing them away from others, respecting the curvature constraint of the Lorentz space.

By embedding classification loss within the hyperbolic geometry, our model leverages its exponential capacity to represent hierarchies and clusters more efficiently than traditional

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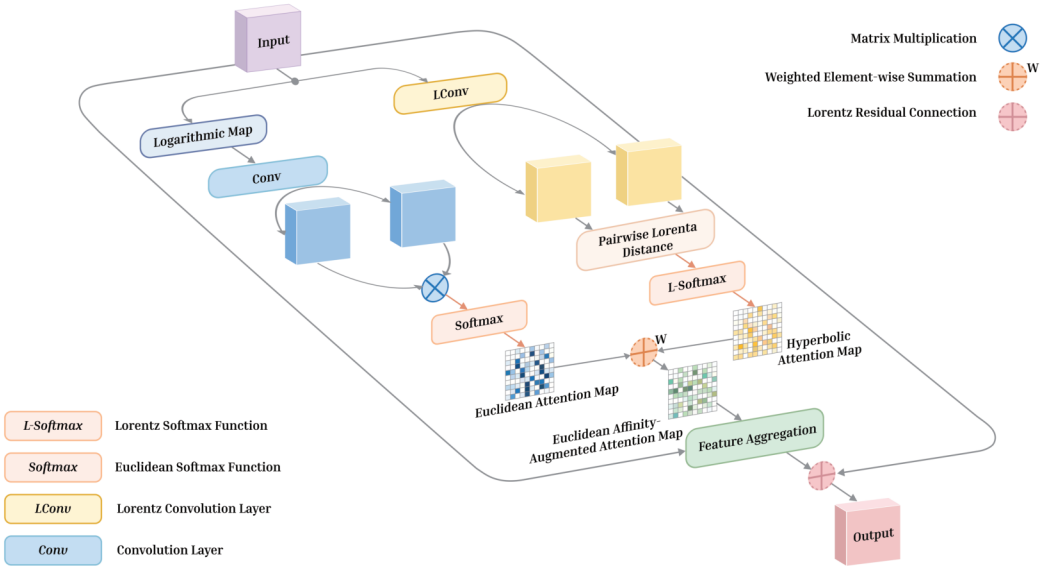


Fig. 5. Details of EAAHAM.

Euclidean softmax. This enhances class separation, particularly for fine-grained boundaries and imbalanced distributions in remote sensing scenarios.

D. Euclidean Affinity-Augmented Hyperbolic Attention Module

To incorporate both Euclidean and non-Euclidean properties of visual representations, we introduce the EAAHAM. As depicted in Fig. 5, this model employs parallel hyperbolic and Euclidean attention branches. The hyperbolic branch initially utilizes an LConv, excluding LBN and LReLU for simplicity. It measures point-wise correlations using Lorentz distance, followed by applying the L-Softmax function to generate a hyperbolic attention map, defined as

$$\mathrm{AttnMap}_L = S\left(d_L\right) = \frac{e^{-d_L(x,x_i)}}{\sum_{j=1}^{n} e^{-d_L\left(x,x_j\right)}} \tag{19}$$

where $e^{-d_L(x,x_i)}$ transforms the Lorentz distance into a similarity measure, with the denominator normalizing these values across all pairs in the input.

Concurrently, the Euclidean branch projects hyperbolic representations into Euclidean space using a logarithmic map, followed by standard self-attention to produce an Euclidean attention map

$$\mathrm{AttnMap}_E = \mathrm{Soft\,max}\left(d_E\right) = \frac{e^{-d_E(x,x_i)}}{\sum_{j=1}^{n} e^{-d_E\left(x,x_j\right)}}. \tag{20}$$

These attention maps are then combined through weighted element-wise summation to form the Euclidean affinity-augmented attention map (EAAAM)

$$\mathrm{AttnMap} = \beta \mathrm{AttnMap}_L + (1-\beta)\,\mathrm{AttnMap}_E \tag{21}$$

where β is a manually set coefficient ($\beta = 0.5$ in this study). Aggregation of scaled feature vectors within the hyperbolic space follows:

$$\mathrm{Agg}\left(x_i\right) = \frac{\sum_j \alpha_{ij} x_j}{\left\|\sum_j \alpha_{ij} x_j\right\|_L} \tag{22}$$

TABLE I
DATASETS

Datasets	Vaihingen	Potsdam	LoveDA
Bands used	R, G, B	R, G, NIR	R, G, B
Spatial resolution	9 cm	5 cm	0.3 m
Number of available images	33	38	4191
Spatial size	About 2494 × 2064	6000 × 6000	1024 × 1024
Imaging sensors	Airborne	Airborne	Satellite
Sub-patch size	256 × 256	256 × 256	256 × 256
Training set (number of images)	14	17	2522
Validation set (number of images)	2	2	834
Test set (number of images)	17	19	835

where $\|\|\|\|_L$ is the Lorentz norm, ensuring that the aggregated vector remains on the hyperboloid

$$\|v\|_L = \sqrt{\left|-v_0^2 + \sum_{k=1}^{n} v_k^2\right|} \tag{23}$$

where v_0 is the time component.

This model enhances the network's ability to process information across different geometrical domains and offers a novel approach to managing data with mixed geometric properties, potentially improving semantic segmentation that requires understanding complex relational and hierarchical structures.

IV. Experiments and Results

A. Datasets

As detailed in Table I, we evaluated our EAAHNet using three datasets.

1) ISPRS Vaihingen Dataset: The ISPRS Vaihingen dataset, derived from an aerial platform, offers very high-resolution imagery at 9 cm per pixel. It comprises 33 images, each with dimensions approximately 2494 × 2064 pixels, and utilizes the red, green, and blue (RGB) spectral bands. The dataset is partitioned into 14 training, two validation, and 17 testing images. The ground-truth labels each pixel with land cover types, designating clutter as the background class.

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TABLE II
EXPERIMENTAL SETTINGS

Items	Settings
Epoch	500
Batch size	64
Initial learning rate	0.02
Learning strategy	Step decay
Momentum	0.9
Optimizaer	Riemannian SGD
GPU typer	Nvidia A40 48GB
Number of used GPUs	2
Hyperbolic curvature K	-1
Loss function	Hyperbolic Softmax cross-entropy

2) ISPRS Potsdam Dataset: The ISPRS Potsdam dataset features images with a higher resolution of 5 cm per pixel. It maintains the same semantic labels as the Vaihingen dataset for consistency. Each of the dataset's images measures 6000 × 6000 pixels and includes red, green, and near-infrared (NIR) bands. The dataset is divided into sets for training (17 images), validation (two images), and testing (19 images).

3) LoveDA Dataset: The LoveDA dataset presents unique challenges for the semantic segmentation of large-scale satellite images with a resolution of 0.3 m. Sourced from Google Earth, it covers more than 536 km^2 across rural and urban regions of Nanjing, Changzhou, and Wuhan. The dataset consists of 1024 × 1024 pixel images, divided into 2522 for training, 834 for validation, and 835 for testing. Its class distribution is notably imbalanced, with significant variation in scale, size, and surface type of congeneric objects, adding complexity to the segmentation task.

B. Experimental Settings

Table II presents the specific experimental settings used for training the proposed EAAHNet. These settings detail the computational framework and hyperparameters chosen to optimize model performance and ensure reproducibility. Two GPUs were used for parallel processing and faster computation. The Riemannian stochastic gradient descent (SGD) optimizer was chosen for its adaptation to the hyperbolic space's geometry by considering the Riemannian manifold's structure. This is essential given the hyperbolic curvature K set at -1, defining the constant negative curvature of the hyperbolic space used in the network's embedding layers. We compared the performance of the proposed EAAHNet against several established semantic segmentation methods, including U-Net [28], DeepLab V3+ [29], and DANet [32]. We also included RSI-specific models such as RAANet [44], MACU-Net [65], SCAttNet [45], A2FPN [66], LANet [43], UNetMamba [71], and BEDSN [70]. Additionally, we evaluated HAENet [21], a model that incorporates hyperbolic attention for semantic segmentation of RSIs. All comparative models were reimplemented in the same computational environment to ensure consistency.

C. Evaluation Metrics

In this research, we use widely accepted metrics to evaluate our model's performance on the test dataset. These metrics include the class-wise F_1 score, the average F_1 score across all classes (AF), overall accuracy (OA), and the mean intersection over union (mIoU). The F_1 score, the harmonic mean of precision and recall, assesses the balance between false positives and false negatives. OA measures the proportion of correctly classified pixels out of the total number of pixels. mIoU, a comprehensive metric, evaluates OA. The formulas for these metrics are provided in the following equations:

$$F_1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} \tag{24}$$

$$\text{OA} = \frac{(\text{TP} + \text{TN})}{(\text{TP} + \text{TN} + \text{FP} + \text{FN})} \tag{25}$$

$$\text{IoU} = \frac{\text{TP}}{(\text{TP} + \text{FP} + \text{FN})} \tag{26}$$

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{27}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{28}$$

where TP, TN, FP, and FN represent the counts of true positives, true negatives, false positives, and false negatives, respectively.

D. Numerical Comparisons With the State-of-the-Art Methods

1) Results on the ISPRS Vaihingen Dataset: Table III presents a comprehensive comparison of EAAHNet against a variety of state-of-the-art models on the ISPRS Vaihingen dataset. EAAHNet achieves the best performance, with the highest mIoU of 82.88%, OA of 89.73%, and AF of 91.92.

For class-wise F_1-scores, EAAHNet achieves the best performance in Building (96.34), Low vegetation (87.01), and Tree (92.60), outperforming all other methods. When compared to recent models from other research groups, such as UNetMamba and BEDSN, EAAHNet continues to show advantages. Although UNetMamba leads in the Car class (91.37) and BEDSN in Impervious surfaces (94.71), EAAHNet remains highly competitive in those categories (89.50 and 94.15, respectively) while leading in most others and in overall metrics.

Earlier baselines such as U-Net and DeepLab V3+ demonstrate lower accuracy across all categories, especially in fine-structured or small-object classes. Advanced architectures like DANet, SCAttNet, and LANet offer improved results but still lag behind EAAHNet in both generalization and fine-grained segmentation.

EAAHNet's success is attributed to the integration of hyperbolic geometry and the EAAHAM module, which together enhance the representation of hierarchical structures and local dependencies. This dual-geometry approach allows the model to simultaneously preserve global context and local detail, resulting in superior segmentation performance on high-resolution RSI.

2) Results on the ISPRS Potsdam Dataset: Table IV provides a comprehensive comparison of EAAHNet with state-of-the-art semantic segmentation methods on the ISPRS Potsdam dataset. Our model achieves the best overall performance, with an mIoU of 84.94%, OA of 92.01%, and AF of 93.97, surpassing all baseline methods.

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TABLE III

RESULTS ON THE ISPRS VAIHINGEN DATASET. CLASS-WISE F_1-SCORE, AF, OA (%), AND MIOU (%) ARE LISTED, WHERE THE BOLD TEXT INDICATES THE BEST RESULTS

Methods	Impervious surfaces	Building	Low vegetation	Tree	Car	AF	OA	mIoU
U-Net [28]	84.67	86.13	69.49	82.71	41.89	72.98	71.24	66.66
DeepLab V3+ [29]	87.99	87.80	72.72	85.55	47.37	76.29	74.47	69.69
DANet [32]	91.57	90.37	80.01	88.15	58.50	81.72	79.78	74.65
RAANet [44]	90.09	89.72	76.22	86.13	48.55	78.14	75.33	70.49
MACU-Net [65]	89.52	90.11	74.32	87.15	62.70	80.76	77.86	72.86
SCAttNet [45]	90.44	91.63	81.22	81.49	71.54	83.26	81.52	75.29
A2FPN [66]	92.99	95.17	80.98	89.25	87.62	89.20	87.07	81.06
LANet [43]	93.81	95.70	82.93	91.94	88.44	90.56	88.48	81.72
UNetMamba [71]	92.47	94.33	84.65	88.12	**91.37**	90.19	88.72	81.55
BEDSN [70]	**94.71**	92.30	83.71	89.24	86.31	89.25	86.93	80.82
HAENet [21]	93.16	95.35	81.13	89.42	87.78	89.37	87.23	81.21
Ours (EAAHNet)	94.15	**96.34**	**87.01**	**92.60**	89.50	**91.92**	**89.73**	**82.88**

TABLE IV

RESULTS ON THE ISPRS POTSDAM DATASET. CLASS-WISE F_1-SCORE, AF, OA (%), AND MIOU (%) ARE LISTED, WHERE THE BOLD TEXT INDICATES THE BEST RESULTS

Methods	Impervious surfaces	Building	Low vegetation	Tree	Car	AF	OA	mIoU
U-Net [28]	87.51	89.32	73.98	87.03	48.19	77.21	75.83	70.02
DeepLab V3+ [29]	84.49	86.13	77.37	77.37	85.47	82.17	80.82	73.59
DANet [32]	86.96	92.04	80.14	79.95	89.36	85.69	83.99	77.13
RAANet [44]	90.40	95.70	87.20	81.80	77.60	86.54	85.21	77.95
MACU-Net [65]	89.16	92.76	86.80	83.02	79.25	86.20	85.90	77.99
SCAttNet [45]	91.87	96.90	85.24	87.05	92.78	90.77	89.06	80.83
A2FPN [66]	90.67	95.69	85.54	85.44	91.55	89.78	88.25	80.67
LANet [43]	92.41	96.38	86.82	87.77	93.44	91.36	90.26	81.19
UNetMamba [71]	92.84	96.27	85.41	86.52	92.93	90.39	88.97	80.29
BEDSN [70]	93.89	**97.93**	86.33	88.09	94.21	92.09	90.71	82.31
HAENet [21]	93.91	97.41	88.11	89.12	95.29	92.77	90.12	84.28
Ours (EAAHNet)	**94.45**	97.50	**89.35**	**91.91**	**96.66**	**93.97**	**92.01**	**84.94**

TABLE V

RESULTS ON THE LOVEDA DATASET. CLASS-WISE F_1-SCORE, AF, OA (%), AND MIOU (%) ARE LISTED, WHERE THE BOLD TEXT INDICATES THE BEST RESULTS

methods	background	Building	Road	Water	Barren	Forest	Agriculture	AF	OA	mIoU
U-Net [28]	50.21	54.74	56.38	77.12	18.09	48.93	66.05	53.07	51.81	47.84
DeepLab V3+ [29]	52.29	54.99	57.16	77.96	16.11	48.18	67.79	53.50	52.30	47.62
DANet [32]	54.47	61.02	63.37	79.17	26.63	52.28	70.02	58.14	54.64	50.18
RAANet [44]	55.02	62.19	65.58	81.03	29.25	54.11	74.07	60.18	58.95	53.93
MACU-Net [65]	59.16	64.08	66.73	81.01	32.23	55.81	75.79	62.12	59.65	54.16
SCAttNet [45]	65.95	71.88	77.04	86.61	50.79	61.19	82.00	70.78	67.31	61.09
A2FPN [66]	65.17	73.32	75.19	88.01	48.82	59.96	79.71	70.03	66.89	61.14
LANet [43]	68.32	75.61	79.02	89.21	53.23	66.02	82.34	73.39	70.43	63.35
UNetMamba [71]	71.94	78.76	83.21	91.64	55.79	**69.13**	85.51	76.57	73.11	66.45
BEDSN [70]	**72.33**	80.18	**84.37**	92.14	**56.62**	69.02	86.47	77.02	73.39	66.71
HAENet [21]	68.06	75.32	78.72	88.87	53.03	65.77	82.02	73.11	70.21	64.17
Ours (EAAHNet)	71.86	**80.54**	83.62	**93.27**	55.90	68.17	**86.33**	**77.10**	**73.77**	**67.12**

For class-wise performance, EAAHNet ranks first across all categories. It achieves F_1-scores of 94.45 for Impervious surfaces, 97.50 for Building, 89.35 for Low vegetation, 91.91 for Tree, and 96.66 for Car. Compared to the recent models UNetMamba and BEDSN, EAAHNet improves in every class, particularly in Low vegetation and Tree, where precise delineation is often challenging due to spectral similarity and structural overlap.

Earlier models such as U-Net and DeepLab V3+ demonstrate significantly lower scores across all metrics, particularly in small object classes like Car. More recent attention-based or dual-stream networks such as DANet, SCAttNet, A2FPN, and LANet achieve strong results, yet still fall short of EAAHNet, especially in AF and mIoU. Although BEDSN and HAENet approach EAAHNet's performance in some categories, their overall results remain lower, indicating the robustness and consistency of our method.

EAAHNet's superior segmentation performance can be attributed to the integration of hyperbolic geometry with Euclidean affinity modeling through EAAHAM, which enables more expressive representation of spatial hierarchies and boundary structures. These strengths are particularly impactful in large-scale, high-resolution datasets like Potsdam, where both global context and fine-grained object detail are critical.

3) Results on the LoveDA Dataset: Table V presents a comparative evaluation of EAAHNet against a range of state-of-the-art models on the LoveDA dataset. Our method achieves

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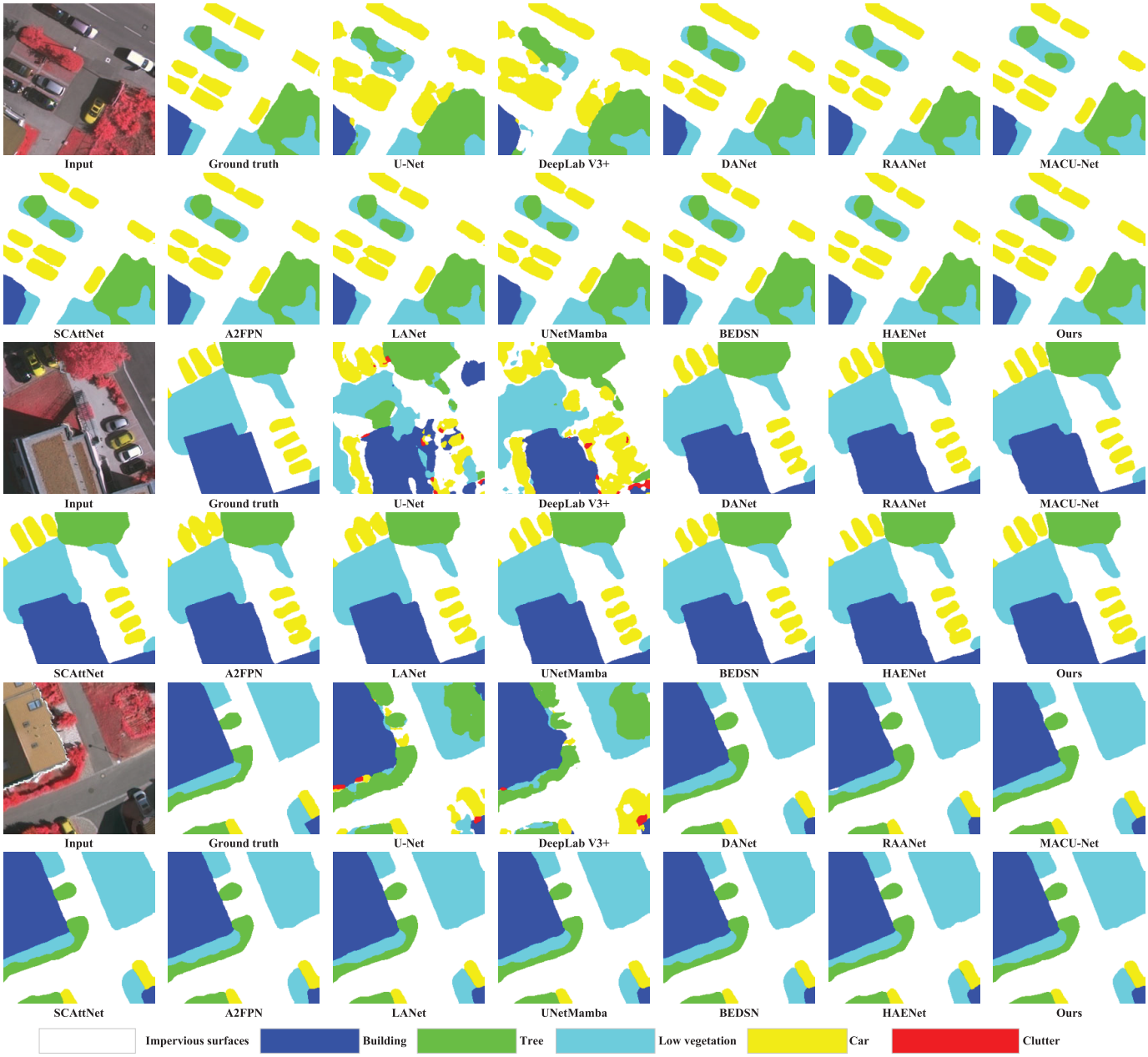


Fig. 6. Visualizations on the ISPRS Vaihingen test set.

the highest mIoU of 67.12%, OA of 73.77%, and AF of 77.10, clearly outperforming all baseline and recent methods.

EAAHNet demonstrates superior performance in class-wise segmentation, achieving the highest F_1-scores in Building (80.54), Water (93.27), and Agriculture (86.33). While BEDSN leads slightly in Background, Road, and Barren classes, EAAHNet still delivers highly competitive scores and ranks second in most of them. Compared to UNetMamba, EAAHNet exhibits consistent improvements in Building, Water, and overall metrics. These results highlight EAAHNet's capacity to accurately delineate both densely structured urban objects and complex natural surfaces.

Earlier methods like U-Net and DeepLab V3+ perform substantially worse, particularly in the Barren and Forest categories, which require fine contextual reasoning. DANet, RAANet, and MACU-Net provide incremental improvements, but lack the global-local structural modeling required for optimal segmentation. Advanced attention-based models such as SCAttNet and A2FPN improve significantly, but still fall short of our method, especially in AF and mIoU.

EAAHNet's strength lies in the joint modeling of hierarchical relationships in hyperbolic space and affinity-guided contextual enhancement from Euclidean space via EAAHAM. This enables more effective learning from complex geospatial patterns, which are prevalent in the LoveDA dataset that contains both rural and urban regions with diverse semantics.

E. Visual Comparison With the State-of-the-Art Methods

1) Visual Inspection on the ISPRS Vaihingen Dataset: To further evaluate the qualitative performance of different

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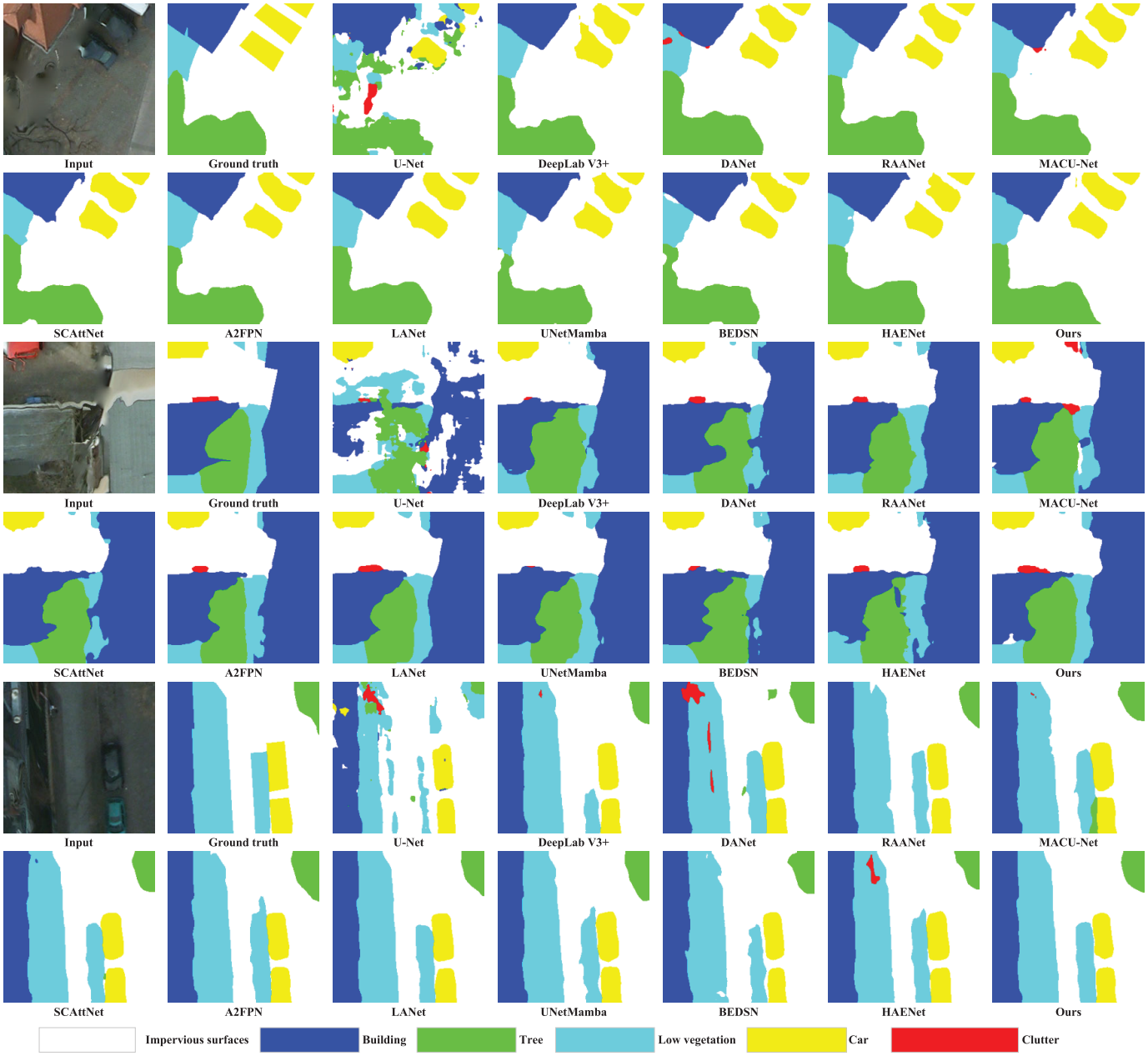


Fig. 7. Visualizations on the ISPRS Potsdam test set.

segmentation models, Fig. 6 presents a visual comparison of representative results on the ISPRS Vaihingen dataset.

As shown in the figure, EAAHNet produces segmentation maps with notably clearer structure boundaries and higher visual coherence compared to other methods. Specifically, EAAHNet achieves superior delineation of narrow features such as roads and small objects like cars, which are often confused by earlier models like U-Net and DeepLab V3+.

Compared to LANet and UNetMamba, our model better preserves class consistency in complex vegetated areas (e.g., low vegetation and tree boundaries) and shows reduced misclassification between spectrally similar categories. While BEDSN exhibits competitive performance, it fails to fully capture the spatial granularity and object contours in regions with dense urban texture.

Overall, the visual inspection confirms that EAAHNet excels in maintaining object boundary integrity and contextual accuracy, demonstrating its strong capability in high-resolution urban scene understanding.

2) Visual Inspection on the ISPRS Potsdam Dataset: To complement the quantitative evaluation, Fig. 7 displays visual comparisons of segmentation results on the ISPRS Potsdam dataset.

The visual results highlight EAAHNet's superior capacity to maintain object boundary integrity, particularly in dense urban scenes where cluttered structures and fine-grained categories co-exist. Our model preserves spatial details and minimizes class confusion across complex categories such as Impervious surfaces, Building, Low vegetation, Tree, and Car.

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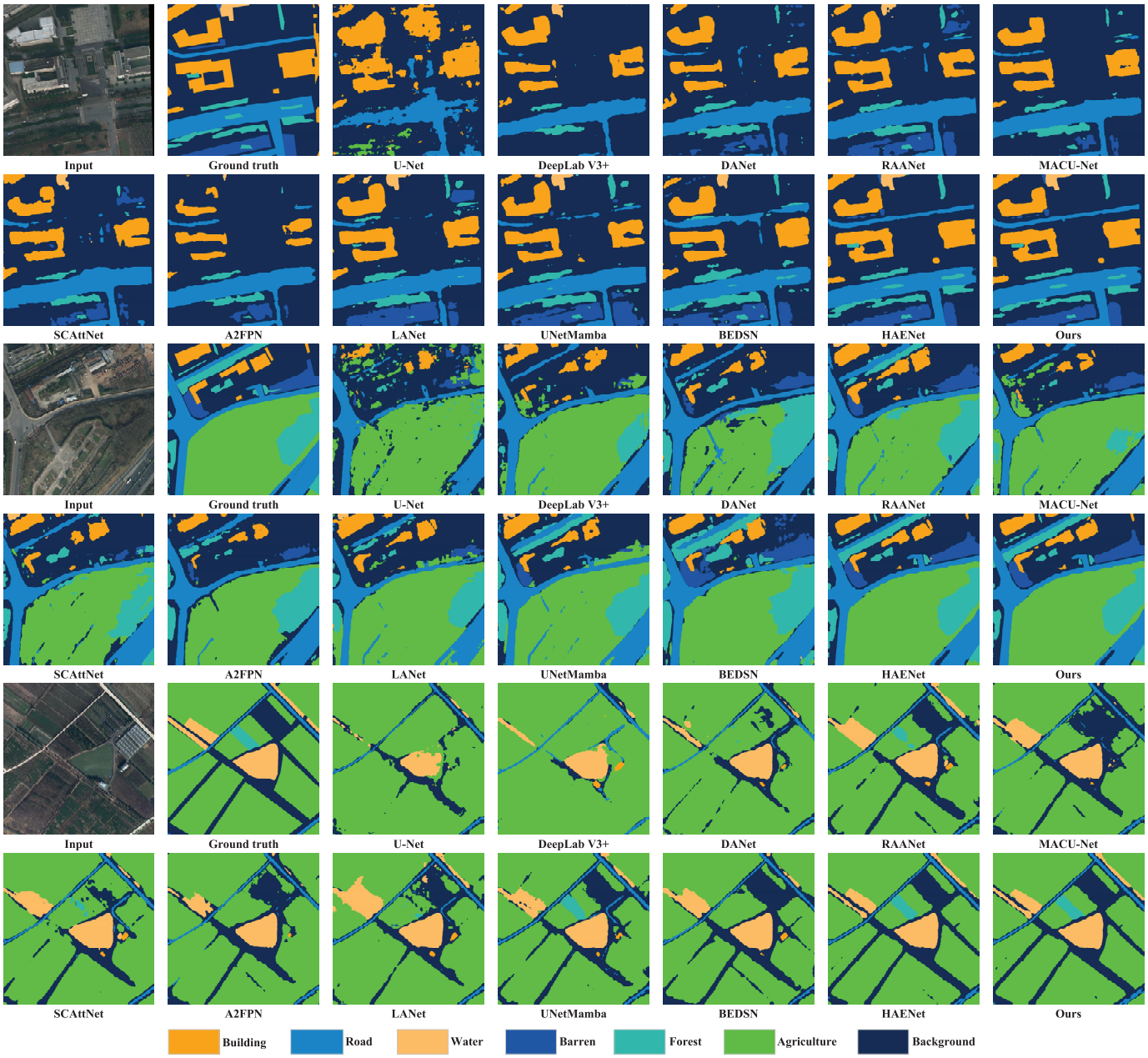


Fig. 8. Visualizations on the LoveDA test set.

Compared with LANet and HAENet, which already produce refined outputs, EAAHNet offers more coherent semantic boundaries and better differentiation between spectrally similar vegetation classes. While UNetMamba and BEDSN perform well in smoother regions, their predictions suffer from structural fragmentation in heterogeneous zones.

Additionally, earlier methods like U-Net, DeepLab V3+, and DANet fail to capture small object instances, such as cars, or produce incomplete edge definitions in boundary regions. By contrast, EAAHNet shows high visual fidelity even in narrow road structures and mixed vegetative areas, underscoring its robust feature representation across diverse classes.

3) Visual Inspection on the LoveDA Dataset: To assess visual segmentation quality under complex geographic and semantic conditions, Fig. 8 provides representative results from the LoveDA dataset.

EAAHNet demonstrates strong visual consistency across diverse environments, including both dense urban areas and unstructured rural regions. In contrast to early models like U-Net and DeepLab V3+, which often fail to distinguish Barren and Forest or confuse Road with Building, EAAHNet produces more structurally coherent maps with fewer fragmented predictions.

Compared with UNetMamba and BEDSN, EAAHNet better preserves boundary continuity and spatial transitions, particularly in challenging categories such as Barren and Forest. While BEDSN performs well in Background and Road, it tends to blur class separations in agriculturally mixed regions. UNetMamba also exhibits localized artifacts near vegetation-water boundaries.

Overall, the visual comparisons highlight EAAHNet's ability to maintain semantic integrity and fine-grained spatial

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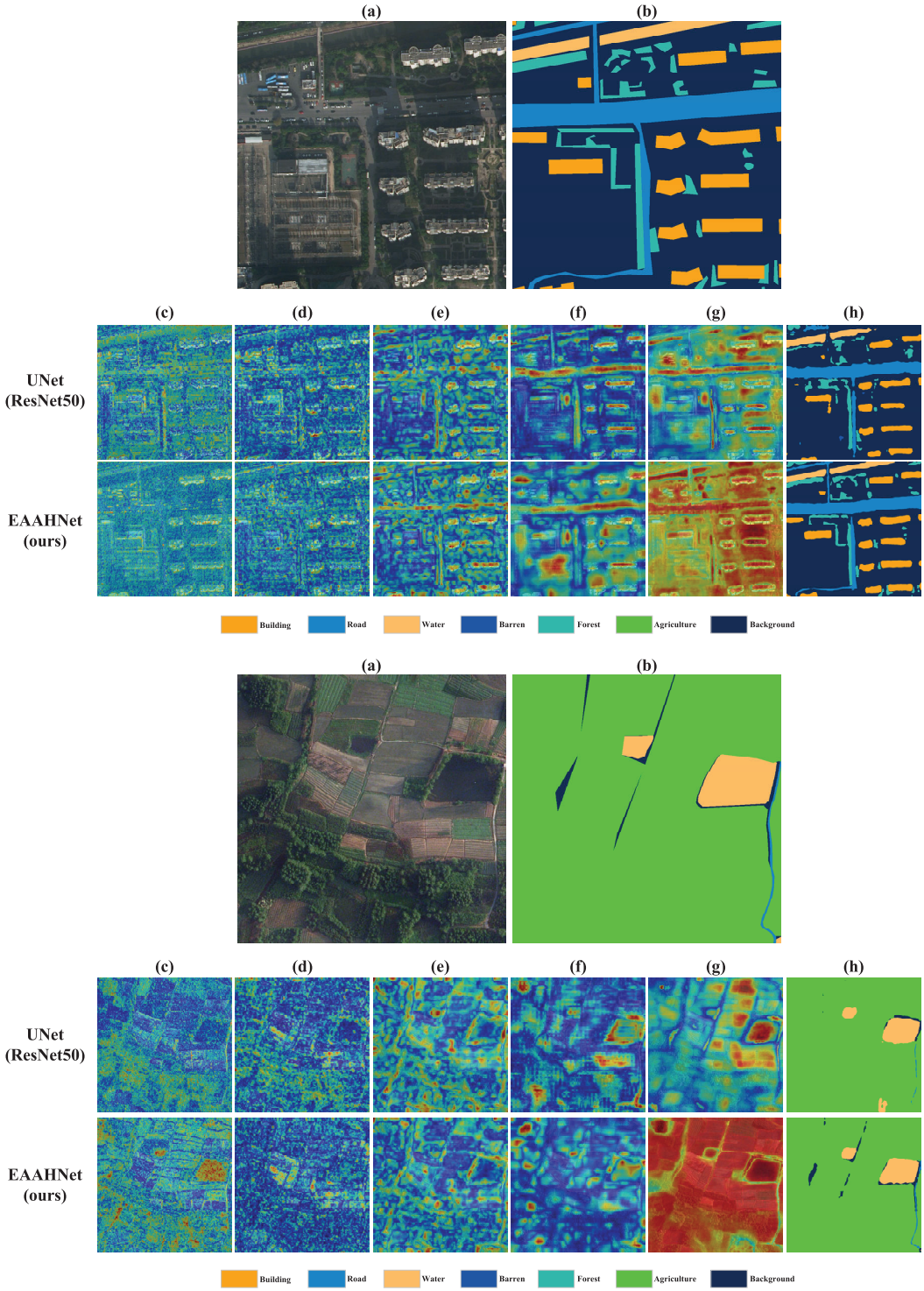


Fig. 9. Feature map visualizations of U-Net (using ResNet 50 as backbone) and EAAHNet (ours) of random samples from the LoveDA test set. (a) Input, (b) ground truth, (c)–(f) feature maps of different stages of feature encoder, (g) feature maps of segmentation head, and (h) predicted results.

structure. Its use of hyperbolic embeddings and affinity-augmented attention enables more robust reasoning over class hierarchies and transitions, making it well-suited for heterogeneous landscapes like those in the LoveDA dataset.

F. Effects of β

To address the concern regarding the choice of $\beta = 0.5$ in (17), we conduct an ablation study to investigate the sensitivity of the model to different β values. This parameter controls the balance between Euclidean affinity and hyperbolic embedding, and its setting is crucial for optimizing the dual-geometry fusion.

We evaluate the performance of our model using various values of $\beta \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ on three datasets: ISPRS Vaihingen, ISPRS Potsdam, and LoveDA. As shown in Table VI, the model achieves the best results consistently when $\beta = 0.5$, supporting our design choice.

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TABLE VI

PERFORMANCE UNDER DIFFERENT β VALUES [IN (21)] ON THREE DATASETS. EACH RESULT IS FORMATTED AS AF/OA/MIOU

β	Vaihingen	Potsdam	LoveDA
0.1	90.91/88.21/81.34	92.34/91.55/84.11	76.32/73.04/66.25
0.3	91.43/89.02/82.02	93.21/91.78/84.57	76.85/73.29/66.89
0.5	**91.92/89.73/82.88**	**93.97/92.01/84.94**	**77.10/73.77/67.12**
0.7	91.51/88.87/82.41	93.44/91.63/84.33	76.69/73.11/66.78
0.9	90.78/88.04/81.52	92.11/90.49/83.06	75.91/72.42/65.92

TABLE VII

IMPACTS OF EAAHAM ON THREE BENCHMARKS. RESULTS ARE IN THE FORM OF AF/OA (%) /MIOU (%)

Models	ISPRS Vaihingen	ISPRS Potsdam	LoveDA
EAAHNet (Standard)	91.92/89.73/82.88	93.97/92.01/84.94	77.10/73.77/67.12
EAAHNet (w/o EAAHAM)	89.96/87.81/81.11	90.30/88.41/81.62	71.85/68.84/62.65

These results indicate that overly skewed fusion weights toward either Euclidean or hyperbolic features (e.g., $\beta = 0.1$ or 0.9) lead to suboptimal performance. Setting β to 0.5 ensures a balanced contribution from both spaces, enhancing model robustness across various scene types.

G. Impacts of EAAHAM

To examine the effectiveness of the EAAHAM, we conducted ablation experiments across three datasets: ISPRS Vaihingen, ISPRS Potsdam, and LoveDA. We compare the complete EAAHNet with a variant where EAAHAM is removed (denoted as EAAHNet w/o EAAHAM). In the ablated model, EAAHAM is replaced by a lightweight baseline composed of LConv, LBN, and LNA, following the residual attention pathway illustrated in Fig. 3.

Table VII reports the performance on all datasets in terms of AF, OA, and mIoU. As shown, the removal of EAAHAM leads to consistent performance degradation across all benchmarks. On ISPRS Vaihingen, AF drops by 1.96, OA by 1.92%, and mIoU by 1.77%. On ISPRS Potsdam, the impact is even more pronounced, with reductions of 3.67 in AF, 3.60% in OA, and 3.32% in mIoU. For the LoveDA dataset, which features heterogeneous rural–urban scenes, removing EAAHAM results in a decline of 4.10 in AF, 3.93% in OA, and 3.57% in mIoU.

These results confirm that EAAHAM plays a critical role in capturing spatial dependencies across Euclidean and hyperbolic manifolds. Its design improves semantic discrimination, particularly at object boundaries and in spectrally ambiguous regions. This fusion of geometric representations is especially beneficial for complex structures such as rooftops and vegetation in urban areas, and class transitions in rural environments.

H. Feature Map Visualization

To intuitively understand how hierarchical spatial semantics are captured by our proposed EAAHNet compared to traditional convolutional architectures, we visualize the intermediate and final feature representations from both networks using Grad-CAM [79]. We provide visual comparisons of feature maps extracted by a traditional convolutional network (U-Net with ResNet-50 backbone) and our proposed EAAHNet. As shown in Fig. 9, two representative samples from the LoveDA test set are randomly selected to illustrate the differences in feature activation and semantic focus between the two methods. For each sample, subfigures (a) and (b) show the input image and the corresponding ground truth, respectively. The first row depicts the feature maps of the U-Net baseline, while the second row shows those from EAAHNet. Columns (c) to (f) visualize the Grad-CAM-based feature activations from four successive stages of the encoder. Column (g) displays the feature map from the segmentation head, and column (h) presents the final segmentation result.

From the visualizations, we observe that U-Net tends to produce diffuse and spatially inconsistent activation maps, especially in deeper layers, which often leads to fragmented predictions in the final output. In contrast, EAAHNet demonstrates stronger localization and structurally coherent attention across all layers. The hyperbolic modeling of hierarchical spatial structures allows our network to maintain both fine-grained local feature fidelity and global contextual awareness. This results in better-defined boundaries, reduced confusion in ambiguous regions, and improved semantic consistency—attributes critical for remote sensing imagery where multiscale and nested object representations frequently occur.

These visual comparisons substantiate the superiority of our approach in learning meaningful and hierarchically aware representations, further supporting the quantitative results and theoretical motivation presented throughout this work.

V. CONCLUSION

The research presented in this article substantiates the effectiveness of the proposed EAAHNet, which synergistically combines Euclidean and hyperbolic learning paradigms to advance the field of semantic segmentation for RSIs. The network's architecture, featuring the novel EAAHAM, addresses the intricacies of urban and rural scenes, delivering enhanced segmentation accuracy across diverse landscapes. Experimental results across multiple benchmarks reveal that EAAHNet outperforms conventional segmentation approaches. This work also highlights the representational strength of hyperbolic geometry in capturing hierarchical spatial relationships common in RSIs, with Lorentz-based modeling offering a compact yet expressive embedding of complex features.

Looking ahead, one promising direction is to explore how hierarchical structures embedded in hyperbolic space could improve model generalization in limited data scenarios. Hyperbolic embeddings may offer more efficient inductive bias and semantic capacity under few-shot or weakly supervised settings, which are common in practical remote sensing tasks. Besides, we plan to systematically investigate the impact of varying convolutional kernel sizes in hyperbolic space, especially within Lorentz-based architectures. Exploring adaptive or dynamically learned kernel sizes may further enhance contextual modeling capabilities in remote sensing scenes.

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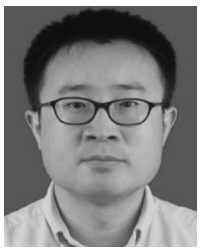


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